

Model Selection and Prediction

A fitted model is judged by how well it predicts responses it has not seen, and its error on the responses it was fitted to flatters it. For any fitting procedure the flattery is twice the covariance between the fit and the data, $2\sigma^2 p/n$ for least squares whether the model is right or wrong. Correcting for it gives Mallows' C_p and AIC. BIC answers a different question, which model is most probable: it finds a true finite model with probability tending to one, while AIC predicts almost as well as the best candidate when no finite model is true. Cross-validation estimates prediction error with fewer assumptions, and for least squares and ridge it needs no refitting. The search over subsets is unstable, and every estimate and interval computed after it is biased by it, unless the inference is designed to survive the search.

What you need. [Chapter 6](#) (projections, Theorem 6.5.1, and the Frisch–Waugh–Lovell theorem, Theorem 6.6.2), [Chapter 7](#) (the likelihood of the normal model, Theorem 7.3.1), [Chapter 9](#) (adjusted R^2 , Proposition 9.5.3), [Chapter 11](#) (the F test, Theorem 11.1.1), [Chapter 13](#) (Scheffé and Bonferroni, Theorem 13.2.2 and Theorem 13.4.1), [Chapter 14](#) (prediction error and its optimism, Proposition 14.2.7 and Theorem 14.2.6), [Chapter 19](#) (omitted and irrelevant regressors, Theorem 19.1.5) and [Chapter 20](#) (the deletion formulas and PRESS, Theorem 20.3.1). From this part: [Chapter 26](#) (collinearity) and [Chapter 27](#) (ridge regression, thresholding and Stein's lemma, Lemma 27.5.1). Every criterion here is an expected sum of squares, so the distribution theory of [Chapter 4](#) is used directly (Theorem 4.3.2, Proposition 4.1.3), with the mean of a quadratic form (Theorem 2.3.1, [Chapter 2](#)) in [Section 29.1](#) and the independence of uncorrelated normal blocks (Theorem 3.4.1, [Chapter 3](#)) in [Section 29.5](#). [Chapter 26](#) is named for the reader who wants to know why a subset can predict better than the full model; no argument here uses it.

29.-1.1 Roadmap

- [Prediction error and optimism](#)
- [Mallows' \$C_p\$, AIC and BIC](#)
- [Cross-validation](#)
- [All subsets and stepwise search](#)
- [Selection bias and honest inference](#)
- [Summary and notes](#)